Structure and a Little Training Beat the Gate: Adaptive-Compute Cascades and Trained Verifiers for Efficient, Accurate Medical VLMs

Li-Wen Kuan (Leo) et al. — conference version (distilled from the long-form manuscript), 2026-07. All numbers verbatim from real checkpoints; none fabricated.

Abstract

Deploying medical vision–language models (VLMs) is expensive: a 32B reasoning model spends ≈ 11 s and ≈ 6 kJ per question (batch-1), a 7B model ≈ 0.2 s. A **cascade** — answer cheaply, escalate only the hard cases — is the standard route to efficiency, but it requires *deciding* which cases to escalate and *what to do* on escalation. We report a hard negative, two positive levers, and a unifying explanation. **The luck floor:** over a *frozen* model, no *training-free* signal beats trivial baselines — a dozen escalation signals hit the same recoverability ceiling (AUROC ≈ 0.5–0.69) because the two models fail together ($\varphi = 0.372$; competent-set P(32B wrong | 7B wrong) = 0.584), and the same floor holds for selection and even bounding boxes. **Two levers move the needle, neither a "better gate."** (1) *Structure:* the Adaptive-Compute Cascade (ACC) routes across compute configurations — 7B-no-think → the 32B's fast no-think mode → 32B-think — exploiting that reasoning over-thinks perception, cutting latency 11.34 s → 2.27 s (−80 %), FLOPs to 52 %, energy ≈ 5× at parity; and a faithful cross-family MCQ margin cascade matches the 32B at −17 … −69 % FLOPs, with a reasoning think tier adding MMMU accuracy (+0.03 … +0.12) across three families. (2) *A little training:* a small trained outcome verifier for best-of-N selection **breaks the luck floor** and **beats the strong 32B/38B on accuracy across three families** (Lingshu 0.421 vs 0.331, MedVLThinker 0.344 vs 0.277, InternVL3 0.255 vs 0.218) and held-out OOD, for free-text answers (35–49 % of the oracle gap) and boxes (SLAKE 40–53 %; the real MS-CXR chest-X-ray benchmark 77–78 %, a 5.6× lift). It discriminates correct candidates at AUROC 0.924, beats a zero-shot 32B verifier despite being 5× smaller, and transfers zero-shot across architectures. We also show the routing ceiling is partly an MCQ artifact (AUROC ≈ 0.6 → ≈ 0.87 open-ended). We release ACC, the verifier, and the full characterization.

Keywords: medical VLM, cascade, test-time scaling, trained verifier, best-of-N, efficiency, grounding.

1. Introduction

A medical VLM maps an image (radiograph, pathology slide, …) and a question to an answer. The strongest models are large reasoning models emitting a `<think>` trace — accurate but slow and energy-hungry (≈ 11 s, ≈ 6 kJ per question at batch-1 for a 32B) versus ≈ 0.2 s for a 7B. A **cascade** — cheap model first, escalate the hard cases — is the standard route to efficiency; its quality hinges on the **gate** (which queries to escalate) and the **action** (what computation to run on escalation). We characterize the limits of the first and find leverage in the second, plus a separate training-based lever for accuracy.

The gate is saturated. A cascade needs *recoverability* — will the strong model fix this error? — not just cheap-model wrongness. This is near-unpredictable from any frozen signal (AUROC ≤ 0.69) because the two models' errors are correlated ($\varphi = 0.372$): on the competent benchmarks 58.4 % of the cheap model's errors the strong model also misses. No decision rule beats a plain confidence threshold.

The leverage is structural (ACC). Instead of a better gate, we change *what escalation runs*. The 32B's fast no-think mode (≈ 0.34 s) is as accurate as or better than its slow think mode on perception VQA (thinking over-thinks perception: 32B-no-think beats 32B-think by +0.085 on SLAKE and +0.077 on VQA-RAD). Inserting 32B-no-think as an intermediate tier — gated by the agreement of the two no-think legs — lets the slow think pass fire on only the ≈ 15 % reasoning residual.

Training breaks the selection floor. Routing *between* models is capacity-bound, but "given N samples from *one* model, which is correct?" admits a *trained* answer: a small LoRA verifier used for best-of-N selection recovers 35–78 % of the oracle gap for answers and boxes, where every training-free selector is luck-floored — and, across three families, it beats the strong model on accuracy.

Open-ended evaluation matters. The pessimistic routing AUROC (≈ 0.6) is partly a multiple-choice artifact: a single A/B/C/D letter is maximally discrete. On open-ended free-text the same confidence signal reaches AUROC ≈ 0.87, so we evaluate selection/verification open-ended.

Contributions. (i) **ACC**, a compute-configuration cascade with large measured latency/energy/FLOPs savings at parity and a per-benchmark safety guarantee. (ii) A **luck-floor characterization**: a dozen training-free escalation/selection signals capped at the recoverability ceiling, explained by error correlation, extended to actions, cross-family peers, the language prior, and structured outputs. (iii) The **open-ended ceiling-break**: the routing signal is a discreteness artifact (MCQ ≈ 0.6 → open ≈ 0.87). (iv) A **trained outcome verifier** that breaks the selection floor and **beats the 32B/38B across three families + held-out OOD**, for free-text answers and bounding boxes (incl. real MS-CXR), bootstrap-significant, a test-time-scaling method that transfers cross-architecture. (v) A **faithful cross-family MCQ**

cascade + think tier, reproducing published numbers (Lingshu-32B MMMU 0.633 = paper 62.3) and matching the 32B at −17 … −69 % FLOPs. (vi) Honest framing: the agreement *gate* is shared with prior cascading; our novelty is the compute-configuration *structure*, the verifier's *application/unification*, and the characterization.

2. Related Work

Efficient cascades / routing. FrugalGPT-style cascades escalate by a post-generation confidence/benefit signal; agreement-based cascading (ABC) escalates on ensemble disagreement; CAR uses confidence-aware routing. Our agreement gate is, by design, in this family — we do not claim it novel; the contribution is the compute-configuration tiering it controls. **Confidence / selective prediction / conformal.** MSP/Chow, entropy, Gini/DOCTOR, split-conformal (CP-Router/LAC), and learned deferral (L2D) are the training-free gate family we benchmark; all cluster at the recoverability ceiling, consistent with Jitkrittum et al. (NeurIPS 2023), who prove confidence-only deferral is fundamentally limited. **Verifiers for best-of-N.** Generative verifiers (GenRM) cast reward modeling as next-token prediction in text; vision-language process reward models rerank reasoning steps; medical generator–verifier pipelines exist for *data synthesis*. We train an *outcome* verifier for *inference-time* best-of-N in medical VQA and unify it across free-text answers and bounding-box grounding — a combination absent from prior work — as a constructive counter to *Verification Mirage* [2605.10850], which concluded self-verification fails in medical VQA and pointed to retrieval, not a trained verifier. Verifier-score-as-gate precedents (CCPS [2505.21772], Self-REF [2410.13284], Kiyani [2602.17633]) do not train an outcome verifier for a medical-VQA cascade. The multiple-choice-vs-open-ended discreteness claim is ours.

3. Setup, Method, and Definitions

Models. Three families, each a cheap 7B/8B and a strong 32B/38B: **Lingshu-7B/32B**, **MedVLThinker-7B/32B**, **InternVL3-8B/38B**. Verifier base: Lingshu-7B; box-verifier base: Qwen2.5-VL-7B.

Data. Seven MCQ benchmarks: PMC-VQA, SLAKE, VQA-RAD, PathVQA, MMMU-medical, MedXpertQA-MM, OmniMedVQA. ALL-6 = all but OmniMed; ALL-5 = ALL-6 − MedXpert (near-chance); COMPETENT-4 = SLAKE/VQA-RAD/PathVQA/PMC. Open-ended sets (LLM-judge graded): SLAKE/VQA-RAD/PathVQA/Kvasir-VQA-x1, + RadImageNet-VQA (transfer/OOD). Grounding: SLAKE organ boxes and the real **MS-CXR** chest-X-ray benchmark (PhysioNet). The MCQ efficiency results use the **faithful MedEvalKit protocol** (the model authors' harness; isolated vLLM 0.9.0.1 + the correct wrappers), under which Lingshu-32B MMMU = **0.633 = paper 62.3** (exact); our internal harness is *not* faithful. The LLM judge (MedVLThinker-32B, cross-checked by a Claude-Sonnet-5 judge) is validated: 100 % exact-match anchor and independent-judge κ = 0.85–0.96.

Cost model. One forward = F = 2·N·(P+G) FLOPs (N_7 = 7.6e9, N_{32} = 33.0e9). A cascade with per-tier cost c and escalation probabilities e_0, e_1 costs C = c_T0 + e_0·c_T1 + e_1·c_T2 (Eq. 1); FLOPs% = Σcascade / Σalways- 32B-think; latency/energy use Eq. 1 with measured batch-1 per-tier costs (energy NVML-integrated). Best-of-K FLOP cost in 7B-forward-equivalents: K = 1 → 1×, 2 → 4×, 4 → 8×, 8 → 16×; always-32B = 4.57×.

Signals, verifier, metrics. Confidence signals from candidate-letter logprobs: margin $m = \square_1 - \square_2$; MSP; entropy; Gini. A gate escalates iff m < τ. The **verifier** s_φ(v,q,a) = P_φ(Yes | v,q,a) = softmax(z)_Yes (Eq. 4) is a LoRA-fine-tuned VLM trained by binary cross-entropy on per-sample correctness labels y (Eq. 5; LLM-judge for answers, y = 1[IoU ≥ 0.3] for boxes); **best-of-N** returns â = argmax_i s_φ. Accuracy = exact-match (MCQ) or LLM-judge (open); AUROC; oracle@N; gap-captured = (acc(â) − greedy)/(oracle@N − greedy) (Eq. 2); guard = # benchmarks worse than always-7B (0 = never worse).

Method I — ACC. A three-tier cascade over compute configurations of the *same* two models: T0 = 7B-no-think @cap320 (≈ 0.21 s) → T1 = 32B-no-think @cap320 (≈ 0.34 s) → T2 = 32B-think @fullres (≈ 11.3 s). T0 outputs iff margin $m_0 \geq \tau_0$. The think tier fires only when the two no-think legs *disagree*, with an ε-margin tiebreak so it triggers on the lowest-margin disagreements (≈ 15 %, not the full ≈ 32 % disagreement rate). The only fitted parameters are the scalars τ_0, τ_1, set on held-out PMC-VQA-train.

Method II — the verifier. Best-of-N selection with Eq. 4. It is *not* circular: the judge is an automated grader of match-to-*gold* (labels from the answer key, not the 32B's knowledge); the verifier trains on the 7B's *own* samples and only *discriminates* correct answers (strictly easier than generating them), which is why a 7B suffices and imports no 32B capability. AUROC 0.924 and a −0.047 blank-image ablation confirm genuine visual discrimination.

4. The Luck Floor

Over a frozen model, no training-free decision rule beats trivial baselines — for the gate, the action, *or* selection.

(a) The gate is saturated. Across 12 signal families (margin, MSP/Chow, entropy, Gini/DOCTOR, energy, hidden-state probe, self-verification P(True), conformal/CP-Router, learned GBM router, cross-model agreement, multi-resolution stability, semantic self-consistency), recoverability AUROC sits at **0.5–0.69** (hidden-state probe 0.60 vs confidence 0.68). The cause is error correlation: ALL-6 recoverability P(32B right | 7B wrong) = 37.2 % (φ = 0.372);

COMPETENT-4 P(32B wrong | 7B wrong) = 0.584 — the models fail together. **(b) Cross-family complementarity is real but unexploitable:** oracle union(7B | InternVL) = 0.753, +Phi = 0.801 vs always-32B 0.645, yet the best learned router (0.621) ≈ always-7B (0.622) and a SigLIP image+text router is at chance. **(c) A language prior:** over half the 7B's answers are unchanged when the image is blanked, and image-sensitive vs insensitive questions have near-equal accuracy (0.620 vs 0.625). **(d) The action axis is capacity-bound:** cheap same-model repairs recover 14.3 % of errors the 32B *also* misses (a novel observation) but are unharvestable — a confidence-gated repair ladder *loses* at parity (43 % vs 39 % compute). **(e) Selection is luck-floored (the headline negative):** sampling one model N = 8 times leaves a large oracle gap (SLAKE-open greedy 0.730 → oracle@8 0.879), but the model cannot say *which* sample is right. Every training-free selector ties or trails random-pick (0.720): self-verify P(Yes) 0.715, 32B pointwise 0.746, 32B listwise 0.758 (best, 24 % of gap), fusion 0.743 — none beats the 32B single pass (0.819); synthesis *backfires* (0.774). A **majority trap** explains it: the correct answer is a minority vote in 74–90 % of recoverable cases. The same holds for **bounding boxes** (SC-medoid at the luck floor).

5. The Open-Ended Ceiling-Break

The pessimistic gate AUROC (≈ 0.6) is inflated-pessimistic by multiple-choice. Re-running the *same* confidence signal on open-ended free-text, detection of cheap-model errors jumps to **AUROC ≈ 0.87** (Lingshu-7B→32B confidence 0.866; pooled 0.846, 95 % CI [0.830, 0.862]) — a *discreteness*, not answer-length, effect (token-F1 grading agrees). The gate itself, however, *remains* near-optimal at plain confidence even here (confidence 0.866 vs exact-SC 0.845, semantic-SC 0.806, P(True) 0.755) — so even in the favorable regime, *training-free* selection is still floored, motivating the trained verifier.

6. Results

6.1 ACC: large efficiency gains at parity, and it is the structure not the gate

At parity accuracy (always-32B-think) on ALL-6 (MedVLThinker, calibration held out):

system	acc	esc$_0$	think	FLOPs%	latency	energy	guard
always-7B-nt @cap320	0.5262	0 %	0 %	8.4 %	0.13 s	20 J	—
always-32B-nt @cap320	0.5573	100 %	0 %	36.2 %	0.23 s	78 J	0.0
always-32B-think [PARITY]	**0.5723**	100 %	100 %	100 %	**11.34 s**	**6319 J**	0.0
Ours: ACC-v2 (agreement)	**0.5693**	71.7 %	15.1 %	**52.0 %**	**2.27 s**	**1182 J**	**0.0**
ACC-v1 (margin)	0.5687	66 %	19.5 %	53.9 %	2.69 s	1417 J	0.0

ALL-6: 11.34 s → 2.27 s (−80 % latency), FLOPs 100 → 52 %, energy ≈ 5× (6.3 kJ → 1.2 kJ), at −0.003 accuracy and guard 0 (never worse than the 7B on any benchmark). On **ALL-5** (excluding near-chance MedXpert): 8.88 s → **0.44 s**, FLOPs to **24.9 %**, energy to 173 J. **It is the structure, not the gate:** holding the 3-tier configuration fixed and swapping in every training-free gate (MSP/Chow, entropy, Gini/DOCTOR, AutoMix self-verify, FrugalGPT-learned, Jitkrittum L2D) lands them all on the same accuracy–FLOPs frontier (FLOPs 49–62 %, latency 1.8–8.0 s); agreement is merely the cheapest point. The premise and savings reproduce across five families/three architectures (e.g. Lingshu ALL-5 FLOPs 77.8 % → 38.9 %); where thinking *hurts* (QoQ-Med, Chiron), ACC gracefully collapses to the cheap 7B (guard 0).

6.2 The verifier beats the 32B across three families

Trained outcome-verifier best-of-8 + verifier-confidence gate, full open-ended sets (InternVL3 uses the *Lingshu-trained* verifier — cross-architecture transfer). "cascade-best" = best point on the accuracy-vs-escalation frontier.

f	dataset	cheap(SC)	STRONG	verifier-bo8	cascade-best @esc	oracle@8
Lingshu	POOLED	0.322	0.331	0.414	**0.421 @12 %**	0.513
(32B)	VQA-RAD	0.465	0.600	0.575	**0.625 @24 %**	0.630
	PathVQA	0.324	0.376	0.453	**0.469 @33 %**	0.517
	RadImageNet-OOD	0.329	0.289	0.353	**0.353 @0 %**	0.512
MedVLThinker	POOLED	0.266	0.277	0.339	**0.344 @14 %**	0.416
(32B)	VQA-RAD	0.420	0.525	0.490	**0.555 @54 %**	0.600
	RadImageNet-OOD	0.204	0.202	0.241	**0.243 @13 %**	0.317
InternVL3	POOLED	0.202	0.218	0.249	**0.255 @36 %**	0.337

(38B x-arch)	VQA-RAD	0.445	0.415	0.570	**0.580 @12 %**	0.620
	RadImageNet-OOD	0.285	0.304	0.302	**0.313 @52 %**	0.398

The cascade beats the strong model on **every family × dataset cell**, including held-out OOD, and the Lingshu verifier **transfers cross-architecture** to lift a 38B InternVL3 it never trained on. The gate does real work where the 32B is genuinely better (pure best-of-8 loses on the VQA-RAD cells; the gate escalates more there and climbs above the 32B by *knowing when to defer*).

A strict same-split test, honestly. Pooling the four Lingshu sets and holding out 30 % (n = 1064): verifier **0.501 > 32B 0.462 > SC 0.411 ≈ greedy 0.413** — the trained 7B captures **49 %** of the oracle gap and, pooled, edges past the 32B while 5× smaller, beating it on the hard sets (PathVQA 0.441 vs 0.377, Kvasir 0.405 vs 0.326) and losing where the 32B is stronger (SLAKE 0.762 vs 0.829) or n is tiny (VQA-RAD). Across two seeds the win vs the 32B is **+0.039 (95 % CI [+0.010, +0.066], significant)** on seed-0 and **−0.005 (a tie)** on seed-1 — so versus the 32B the honest claim is **matches / modest-win (mean +0.017)**, while the margin over *training-free* selection is robust (+0.088 / +0.066 over greedy; bootstrap vs first-sample +0.116 [+0.092, +0.140]), gap-captured **35–49 %**.

Training is the active ingredient; candidate quality is the ceiling. Zero-shot self-verification is luck-floored; the trained 7B verifier beats a **zero-shot 32B** verifier (0.403 vs 0.355), argmax beats any vote-flavored rule (0.501 > weighted 0.489 > hybrid 0.470), and best-of-K rises monotonically with the budget (K = 1,2,4,8: 0.385 → 0.425 → 0.476 → 0.501; random flat ≈ 0.39) — a genuine test-time-scaling method. It transfers zero-shot to a fifth dataset (RadImageNet 0.329 → 0.353) and across generators (Lingshu verifier → MedVLThinker answers: SLAKE 49 %, VQA-RAD 61 %), though a from-scratch MedVLThinker verifier is weaker (SLAKE 42 %, pooled 25 %, and *fails* VQA-RAD at n = 54 — an honest negative). The binding limit is *candidate quality*: selection efficiency is 74–82 % and a ranking loss lifts per-answer AUROC 0.90 → 0.93 without moving selection, whereas a **cross-model candidate pool raises oracle@N by +0.11–0.15**.

Structured outputs. A LoRA box-verifier selecting best-of-8 boxes recovers **40–53 %** (SLAKE organs) and **77–78 %** of the oracle gap on the real **MS-CXR** chest-X-ray benchmark (gain **+0.191, 95 % CI [+0.152, +0.232]**, n = 435, a 5.6× lift over greedy) — both training-free selectors sit at the luck floor.

The gate cannot be improved. Verifier-confidence is the best cascade gate (pick-correctness AUROC 0.853 / 0.885 / 0.875 across families); the best *trained* gate (GBM/MLP on verifier±cheap features) adds ≤ +0.008 (noise) and the SOTA post-hoc recoverability gate (Jitkrittum Diff-Prob, AUROC 0.708) is worse — because the recoverability wall persists (the strong model fixes only 6–10 % of the verifier's errors, 26 % where it is competent, and *which* is near-unlearnable).

6.3 The faithful cross-family MCQ cascade and a reasoning think tier

Under the faithful MedEvalKit protocol, the 2-tier margin cascade matches always-32B accuracy at large FLOPs savings wherever the small model is competitive (FLOPs saved vs always-32B; keep-cheap / no-win noted):

benchmark	Lingshu	MedVLThinker	InternVL3
MMMU-Med	keep-cheap (7B anomaly)	−14 %	**−62 %**
PMC-VQA (33k)	**−69 %**	−49 %	−16 %
SLAKE	−56 %	no win (7B weak)	keep-cheap (8B ≥ 38B)
VQA-RAD	−17 %	−41 %	**−67 %**
PathVQA	−31 %	**−68 %**	−20 %
MedXpert-MM	no win (floor)	no win	*(in progress)*

Win magnitude ≈ the (32B − 7B) accuracy gap; a held-out-τ honesty check keeps PMC-VQA at −57 % (Lingshu) / −49 % (MedVLThinker) with a fair threshold. A 3-tier **think tier** (adding 32B-think for reasoning) then raises accuracy on the reasoning benchmark MMMU across **all three** families — **Lingshu +0.034 (0.633 → 0.667), MedVLThinker +0.107 (0.613 → 0.720), InternVL3 +0.120 (0.633 → 0.753)** — and MedVLThinker also gains on MedXpert (+0.045); it is regime-adaptive (think where there is headroom, not at the floor). All three families reason on a *generic* "reason step by step" prompt (gen_toks 3 → 275/561/368; MedVLThinker gains +0.107 while emitting *zero* `<think>` tags), correcting an earlier "no promptable think mode" claim.

7. Discussion and Limitations

Novelty, honestly. ACC's agreement gate is the ABC family and the verifier's mechanism follows GenRM; the contributions are the *compute-configuration structure*, the verifier's *application* + *unification* over answers and boxes, and the *characterization* (the luck floor + the open-ended ceiling-break). **Scope.** The over-thinking premise holds on *perception* (COMPETENT-4); MMMU is competent-but-reasoning (thinking helps, so the ACC over-thinking mechanism does not apply, but the think tier does); MedXpert is near-chance and excluded from headline efficiency. Versus the 32B the verifier is matches-to-modest-win (robust only vs training-free selection); the per-dataset spread is wide

(SLAKE 15 % → Kvasir 58 %); a from-scratch verifier is not uniformly robust; the box-verifier lifts *selection over a frozen generator*, not a trained SOTA grounder. Latency/energy are calibrated batch-1 (Eq. 1); FLOPs are exact. **In progress:** the 7th benchmark (OmniMedVQA, ≈ 89k) is still running, InternVL3-MedXpert awaits a context-length fix, and some July latency was measured under GPU contention and is re-measured serially (accuracy/FLOPs unaffected). Candidate quality is the ceiling the verifier now points to — the highest-value next lever.

8. Conclusion

For efficient, accurate medical VLMs, the routing/selection decision cannot be out-engineered with training-free signals — a dozen hit the same recoverability ceiling, selection sits at a luck floor, and the two frozen models fail together. Two levers give large, real gains: **structure** — ACC cuts latency −80 %, FLOPs to ~½, energy ~5× at parity, and a faithful cross-family MCQ cascade matches the 32B at −17 … −69 % FLOPs with a reasoning think tier — and **a little training** — an outcome verifier that breaks the selection luck floor and **beats the 32B/38B across three families + OOD**, for answers and boxes (incl. real MS-CXR), transferring cross-architecture. We also show the routing ceiling is partly an MCQ artifact (≈ 0.6 → ≈ 0.87 open-ended), so medical-VLM cascades should be evaluated open-ended. We release ACC, the trained verifier, and the full negative-result characterization. *All numbers trace to a checkpoint; none are fabricated.*